

What Scale Buys in Tool-Use Agents, and How to Buy It Cheaply: A Capability×Scale×Lever×Cost Map Where Compliance Is Scale-Invariant

Abstract

Tool-use agents map a natural-language request to a sequence of tool calls under a domain policy, and are usually improved by scaling the base model. We ask instead, function by function and across scale, what scaling actually buys: which sub-capabilities improve with size, which remain limited at every scale, which are better supplied by deterministic scaffolding or by per-domain authored relations (which we call A2), and how these should be assembled to minimize total deployment cost. Treating four benchmarks (τ^2 -bench, SOPBench, TaskBench, and Synth) as measurement instruments, we decompose tool-use failure into *operand execution*, *orchestration load*, and *compliance*, and chart each across model scale (7B–32B measured; 72B and above projected) and across levers (base model, capability scaffold, guarantee scaffold, A2, and learning).

We report three findings. First, operand execution saturates early: given an explicit specification, a 32B base model selects the correct action on 88 of 88 cases, so this is not a learnable gap at that scale. Second, the load dimensions that predict failure retire with scale in a definite order—interference and state-multiplicity bind at 7B and 14B but disappear by 32B, while context length and conditional depth persist—and a controlled probe overturns an interference signal that appears only in observational correlations. Third, compliance is scale-invariant: task success rises with scale (0.19 at 7B to 0.55 at 32B), yet the per-write-opportunity rate of policy-semantic violations does not fall (≈ 0.07 – 0.10 across all measured scales, with overlapping 95% confidence intervals), even as authorization-type violations collapse; a deterministic policy gate eliminates the residual at every scale.

We interpret deterministic scaffolding as a load-reduction operator and fold the map into a total-lifecycle cost objective. On τ^2 -retail, a 32B int8 base with a deterministic gate runs on-premises at roughly an order of magnitude lower cost per request than a frontier API at near-equal compliance, and a small-on-premises-plus-escalation fleet is projected to reach equal-or-better compliance at lower cost. We conclude that scale buys capability but not guarantee: even a frontier model requires a deterministic guarantee layer, so the cost-optimal design is not the smallest model but the total-cost knee—a small base, a capability scaffold that recedes as scale grows, a scale-invariant guarantee scaffold, authored A2, and learning reserved for a small, scale-dependent residual.

1. Introduction

A tool-use agent maps a natural-language request to a sequence of tool calls that satisfy a domain policy. The default recipe for improving such an agent is a larger base model. But "larger" conflates distinct capabilities, and it is expensive on two permanent axes: inference cost (a model twice the size costs more than twice as much per request and is slower, for the lifetime of the deployment) and, for on-premises or sovereign deployment, hardware capital expenditure, since the operator must buy the memory, power, and cooling. The practical question is therefore not *how large a model* but *which capabilities must scale, which can be delegated to cheap deterministic structure, which no model can be trusted to perform, and what mixture minimizes total cost*.

Two deployment realities motivate this study. First, regulated and closed deployments force small on-premises models and make compliance a first-class concern. In regulated or sovereign settings—finance, healthcare, government—customer data cannot leave the premises, so a frontier API is unavailable and the operator runs, and buys the hardware for, a local model. At the same time, regulation increasingly demands traceability, validation, and effective human oversight (for example EU AI Act Articles 12 and 14, and US banking guidance SR 26-2)—but not determinism as such; no major framework textually mandates it, the medical-device provision EU MDR Annex I §17.1 being the lone exception. A deterministic policy gate satisfies these obligations by construction, and under SR 26-2 deterministic rule-based processes fall outside the model-risk perimeter entirely, whereas a stochastic agent can demonstrate compliance only statistically, per sample, with its effective policy unverifiable in advance. Our central finding—that compliance is scale-invariant (§5.3)—implies that such a gate is needed at every scale, frontier models included.

Second, per-domain authoring is the recurring cost, so a fixed scaffold combined with cheap-to-author relations is the path to many domains. Domain-specific learning is slow and expensive and must be redone for each field; if the per-domain artifact (the

A2) can be authored cheaply and the scaffold and learned skill are held fixed, then moving to a new domain costs an A2-swap rather than a retrain. This paper measures single-domain compliance and cost; the multi-domain payoff is the generalization axis of the cost framework (§6), which we validate only partially here (a SOPBench cross-domain check) and otherwise treat as a design premise rather than a measured multi-benchmark result. Taken together, a deployer must use a small on-premises model for reasons of cost and sovereignty, needs a form of compliance that scale does not provide, and wants to amortize effort across domains; the map developed here speaks to all three.

We answer with a capability $\times$ scale $\times$ lever $\times$ cost decomposition. For each functional sub-capability we ask, across scale, whether it (a) grows with scale, (b) is a genuine model limit, (c) is covered by deterministic *scaffold*, or (d) is covered by per-domain authored relations (A2); we then assemble the per-capability levers under a total-lifecycle cost objective and locate the cost knee. The benchmarks are instruments for filling these cells, not the object of study.

Objective. We seek to minimize, simultaneously, the cost of cheap GPU tier and count, the memory footprint (favoring int8 or int4 quantization), the maintenance burden per change, the number of human experts required, and the cost of generalizing to each new field (ideally an A2-swap alone), together with hardware capital and inference operating expenditure. These terms are coupled and point in the same direction: cheap, low-memory hardware favors a small, quantized model, while low maintenance and onboarding favor maximizing the fixed domain-general component (deterministic scaffold plus transferable, forgetting-free learning) and minimizing the per-domain A2. The A2 is the common adversary, since it drives maintenance, experts, and per-field cost, and is worst on-premises where a frontier A2-generator is unavailable; a fixed scaffold and transferable skill are the common ally. The optimum is the total-cost knee, not the smallest model.

Contributions. 1. A controlled, frontier-API-free methodology that isolates operand execution, orchestration load, and compliance across scale (§4), using isolated single-shot probes, existing multi-scale runs, and a local user simulator in place of paid full rollouts. 2. Evidence that operand execution is scale-saturated and not a learnable gap at 32B (§5.1); the apparent gap is criterion interpretation (deterministic computation) and join/disambiguation (addressable by presenting the relevant records). Faithful-formalization fine-tuning is therefore unwarranted at 32B, though a residual may exist at 7B (§7). 3. A load decomposition with a scale response (§5.2): the set of load dimensions that bind is itself scale-dependent, with scale retiring interference and state before length and conditional depth. Controlled generation overturns a correlation visible only in observational data. 4. The finding that compliance is scale-invariant (§5.3): capability rises with scale while policy-semantic violation does not, and a deterministic gate closes the residual at every scale. This is the empirical backbone of a two-kind scaffold—a capability scaffold that recedes with scale, and a guarantee scaffold that does not. 5. A total-cost-optimal lever allocation with a deployment payoff (§5.6 and §6): decidability-first routing, an on-premises cost-per-request advantage of roughly an order of magnitude (we report the exact multiple as an estimate, pending measured token accounting), and a projected fleet design, framed by the cost knee. 6. A proposed unifying operator—deterministic scaffold as load reduction (§3)—for which we establish the components but defer the full predictive test (§7); we therefore present it as a framework hypothesis rather than a verified law.

Our claim is deliberately not "small beats large," a result already established by reinforcement-learned routing systems (§2). It is *which* parts of the gap are scale-invariant and must be offloaded, and a cost-optimal map indicating which lever a deployer should pull for each capability.

2. Related Work

The contribution sits at the intersection of several established lines of work. The central lineage is conceptual—why a language model is good at language yet must offload deep procedures and cannot self-guarantee compliance—rather than the recent "small-beats-large" routers (§2.5).

Language is shallow-parallel; deep procedures must be offloaded. A bounded-depth Transformer forward pass is asymptotically uniform TC■ (Merrill & Sabharwal, 2023); chain-of-thought escapes this only by serializing depth (Feng et al., 2023; Li et al., 2024). Natural language embeds procedures of definite computational class—van Benthem's semantic automata (the quantifier "most" is pushdown), which are cognitively real; Relevance Theory's procedural/conceptual distinction; Goodman's notationality, which concerns cost rather than computability (Zhang & Norman). Thus *classifying* which deep procedure a request embeds is shallow and learnable, whereas *executing* it is serial-deep and best offloaded—the basis of our capability/lever routing.

Cognitive architectures, non-introspective decidability, and scale-invariant epistemic limits. Our system instantiates the CoALA umbrella (Sumers et al., 2024), making concrete what it leaves general: a relational-algebra deterministic core,

empty-relation abstention, and zero-retrain A2-swap. SOAR (Newell, Laird, and Rosenbloom; Laird 2012; Wray, Kirk, and Laird 2025) surfaces indecision architecturally—an *impasse* arising from an absent preference rather than from introspection—which is the precedent for our pattern of mapping uncertainty to an observable empty or ambiguous relation and then asking. Deterministic-first with language-model fallback is established (Zhu & Simmons 2024; NL2GenSym 2025). We add a formalization-faithfulness axis, a *learned* empty-to-ask decision, and a small weight-learned domain-general skill with A2-swap. On epistemics, hallucination is provably scale-invariant (Xu et al. 2024; Kalai et al. 2025, who show that scoring rewards guessing over abstention); our §5.3 is the policy-enforcement analogue—no model self-guarantees compliance, just as none self-guarantees truthfulness, and both demand an external deterministic mechanism.

Language model proposes, deterministic component executes. LLM-Modulo (Kambhampati et al. 2024), PAL (Gao et al. 2023), natural-language-to-PDDL formalizers, and reinforcement learning from verifiable rewards with deterministic checkers (Tülu 3, Lambert et al. 2024) all follow this pattern. The boundary is honest: offload does not always win, owing to formalization overhead on simple cases and to specification leakage, and we treat that boundary as itself measurable.

Guarantees and shielding. Safe-reinforcement-learning shielding (Alshiekh et al. 2018; Jansen et al. 2018) and predictive safety filters (Wabersich & Zeilinger 2021), and their agent-level instances ShieldAgent (2025), AgentSpec (2025), and Formal-LLM (2024), veto actions using hand-written or probabilistic specifications. Our gate differs in being deterministic, in *driving* (repairing) preconditions rather than only vetoing, and in pairing with a learned proposer and zero-retrain transfer.

Learned routing and small-model orchestration (the closest rivals). Routing is a learned, transferable skill (A2FM, xRouter, When2Call); tool-necessity is even linearly decodable from hidden states (AUROC 0.89–0.96). ToolOrchestra (2025) reinforcement-learns an 8B router that delegates to a frontier model, reaching 80.2% at 10.3¢ versus GPT-5’s 77.7% at 31.3¢ on τ^2 —the strongest cost result to date, but one that relies on frontier delegation and is therefore disqualified in the on-premises, no-egress regime; it optimizes cost given black-box components and offers no deterministic offload, by-construction compliance, decidable-versus-learned accounting, or zero-retrain transfer. TRUST (2026) aligns tool-calling decisions to model uncertainty by reinforcement learning but remains black-box. ReDAct (2026) and unified routing-cascading (2024) perform calibrated deferral; our fleet is a deterministic-router instance of the same idea. These works establish that a cost-favorable small-model result exists, which is precisely why our contribution is the analysis and method—what is scale-invariant, which lever fits each capability, and where the cost knee lies—rather than the existence result itself.

Cost, regulation, and industry. Against a pure-scale reading of the Bitter Lesson, Brooks’ “A Better Lesson” (2019) and Chollet (2019) argue that total cost and priors must be counted; our knee operationalizes this. The buyer value of determinism is verifiability and model-risk exemption (SR 26-2; EU AI Act 2024/1689; EU MDR Annex I §17.1 on repeatability). Palantir Foundry, whose Ontology resembles scaffold plus A2 and whose AIP language-model layer resembles a small model with a transferable skill, is the industrial baseline; we differentiate on cost—hardware, experts, configuration-only field-crossing, and zero egress—rather than on functionality.

Load and benchmarks. Our load framing defers to cognitive-load theory and to the long-context “lost in the middle” phenomenon; the novelty is only the combination of load decomposition with scaffold-as-load-reduction for tool use. We use benchmarks as instruments: τ^2 -bench (2025; the τ -bench predecessor, 2024), SOPBench (2025), TaskBench, and Synth/CFB (2025); a precedent for measuring a capability floor in this way is the “Sufficient Context” study (Joren et al., ICLR 2025).

The open intersection. Prior work supplies each piece—procedural semantics and automata, Transformer expressivity, SOAR and CoALA, hallucination inevitability, formalize-and-offload, learned routing, and shielding. The unoccupied intersection is the empirical capability $\times$ scale $\times$ lever $\times$ cost map: which capabilities scale retires, which are genuine scale-invariant limits (compliance, measured here), which lever covers each, and the total-cost-optimal assembly. Recent routers optimize cost over black boxes; we open the box.

3. The Capability $\times$ Scale $\times$ Lever $\times$ Cost Framework

We decompose a tool-use trajectory into typed functional steps. For each function f we populate a row indexed by scale N , by lever $\blacksquare \in \{\text{base, capability scaffold, guarantee scaffold, A2, learning}\}$, and by assembled cost. The framework rests on three structural claims.

First, operand execution, orchestration, and compliance fail for different reasons and respond to different levers: operand execution is atomic execution given a criterion; orchestration is holding interdependent steps coherent under load; compliance is

obeying policy with a guarantee.

Second, lever routing is decidability-first, short-circuiting in order of ascending cost: if a step is decidable from schema, policy, and state, use a scaffold; otherwise, if it requires a domain fact, use A2; otherwise, if it is promptable at the available size, prompt; otherwise, if it is scale-emergent and affordable, scale or replicate cheaply; otherwise, for an irreducible natural-language-to-formal step, use a minimal transferable adapter as a last resort. This inverts the default "scale it" reflex.

Third, there are two kinds of scaffold. A *capability scaffold* (presenting records, computing a value, resolving a reference) substitutes for a missing skill and recedes as scale grows—the F3, raw-pass quantity. A *guarantee scaffold* (policy gates) supplies a property that no probabilistic model provides and is scale-invariant—the F4, compliant-pass quantity. Section 5.3 measures this split, which is why even a frontier system retains a deterministic trust layer.

Load as a vector. Orchestration load is not scalar. From failure forensics we distinguish L_{len} (context to retain), L_{state} (interdependent state), L_{branch} (conditional depth), L_{interf} (confusable entities), and L_{contra} (mid-stream revision). Crucially, load is orthogonal to operand difficulty, so load can be raised with operand difficulty fixed.

Scaffold as a load-reduction operator. Each scaffold removes a load dimension: presenting or autofetching records lowers L_{len} and L_{interf} ; computing a value lowers L_{state} ; gates lower L_{branch} ; a controller lowers L_{state} and L_{len} . This yields the relation $L_{eff} = L - \Delta L_{scaffold}$ and the prediction that f passes at scale N if and only if $L_{eff}(f) < L^*(N)$, where $L^*(N)$ is the load tolerance at scale N . To avoid tautology, ΔL should be estimated independently—from the mechanical feature reduction the scaffold performs—and shown to predict the shift in failure onset rather than read off it. We establish the components of this operator here and defer the full predictive test above 32B to future work (§7), so we present it as a hypothesis rather than a validated law.

4. Method

Benchmarks as instruments. τ^2 -retail is the primary benchmark; SOPBench, TaskBench, and Synth serve as the training substrate for the lever allocation, while τ^2 is held out as a transfer target and is never trained on. The scale axis is Qwen2.5 at 7B, 14B, and 32B (GPTQ int8), all measured; 72B and 235B are projected and discussed in §7.

Cost discipline. The only paid component of the evaluation is a GPT-4.1 user simulator, which we avoid during discovery. Capability is measured with isolated single-shot probes and with existing multi-scale runs; a local user simulator (faithful scripted turns over a local base model) replaces the paid simulator for full-flow checks. Paid full runs are used only as a final confirmation of a conclusion already reached without them, at minimal scope.

Conditions. We compare a *floor* condition (no scaffold) against capability- and guarantee-scaffold conditions (*g15*, *present*, *nested*, *assembled*), and we separate operand execution from criterion interpretation with a *given-spec* versus *goal* manipulation.

Metrics. We report a robust pass rate (pass over all repeated trials per item; because the user-simulator noise is approximately 0.11, single-trial figures are not reported as findings), a raw task-pass rate F3, and a compliant-pass rate F4 (F3 less policy violations). We also report, for each load feature, the partial correlation of failure with that feature controlling for operand difficulty; controlled accuracy-versus-load onset curves; and a cost row per cell (latency, tool round-trips, memory and GPU tier, and cost per request).

5. Results

5.1 Operand execution is scale-saturated, not a learnable gap at 32B

Varying only the user-simulator input, we find that given an explicit option specification (the *given-spec* condition) the 32B base selects the correct variant on 88 of 88 cases (100%); given only the goal (the *goal* condition) it succeeds on 62 of 88 (70%). The 30-point gap is criterion interpretation—argmax or argmin computation, multi-attribute reasoning, and conversational fidelity—rather than execution. Genuine join and disambiguation, where the user describes rather than names the target order, succeeds on 7 of 13 (54%) and is addressable by presenting the candidate records. Operand execution is therefore not a 32B capability gap, and faithful-formalization fine-tuning is unwarranted at that scale because there is no residual to learn. (Every earlier "operand failure" in our forensics dissolved into a measurement artifact—a computation bug, a premature case split, or a probe that supplied the identifier—until the controlled probe read 100%. The corollary that matters for cost is that at 7B the base

may be genuinely weak even given a specification, so the learning lever can re-acquire a role at small scale; see §7.)

A complementary limit is that selection over a large candidate set does not scale in-head: comparative selection at 50 candidates scores 0.02 and ranking at 50 scores approximately 0.30 even including a 235B model, whereas a deterministic engine scores 1.00. Choosing the best among many is best offloaded at any scale. (The 235B figure appears only in this large-candidate selection probe; the scale-axis compliance and load runs span 7B–32B, with 72B and 235B projected.)

5.2 Capability-type load retires with scale, in a definite order

The partial correlation of *floor* failure with each load feature, controlling for operand difficulty, is as follows.

dimension	7B	14B	32B
L_len (length)	+0.20	+0.33	+0.37
L_state	+0.24	+0.21	+0.06
L_branch (conditional)	+0.20	+0.13	+0.26
L_interf	+0.30	+0.20	+0.08
L_contra	+0.06	+0.13	+0.04

The binding set is scale-dependent: at 7B and 14B four dimensions predict failure, but by 32B only length and conditional depth survive, with interference and state retiring first. *L_contra* is too sparse in τ^2 to fit reliably. A controlled probe is more informative than the observational correlation: fixing the operand and varying only the number of confusable variants does not reproduce the strong 7B interference signal.

N	1	2	4	8	16
32B	1.00	1.00	1.00	0.97	0.87
7B	1.00	1.00	1.00	0.90	0.80

The gap at N = 16 is only 0.80 versus 0.87, indicating that the observational interference correlation was largely a size confound. (Single-shot probes under-test the inherently multi-turn dimensions *L_state* and *L_contra*; controlled multi-turn probes are left to future work.)

The residual after these load dimensions is orchestration rather than operand execution. Re-running the robust fail-all set under faithful local turns, 2 of 10 failures are simulator-noise flips, and the dominant residual is orchestration under load: batching that violates a precondition, multi-order tracking that drops an order and fabricates an identifier, conditional sequencing, and context overflow. An isolation plan probe—eliciting the abstract plan with all reads supplied and grading only structure—yields correct structure on 6 of 10, indicating that roughly half of the orchestration residual is execution load (closable by separating planning from execution deterministically) and roughly half is an in-isolation planning miss (the candidate for a learned path-selection lever, which we leave to future work). The ceiling of the assembled stack is a robust pass rate of 0.402.

5.3 Compliance is scale-invariant — the central finding

Separating F3 (task pass) from F4 (compliant pass) on the *floor* condition (no gate) gives the following.

scale	F3	F4	gap	violations (g1 auth / g2 policy)	total viol. %
7B	0.189	0.130	0.059	57 / 31	33.6%
14B	0.468	0.404	0.064	38 / 38	26.3%
32B	0.547	0.491	0.056	6 / 41	14.3%

The absolute F3–F4 gap is roughly flat (0.059, 0.064, 0.056), and policy-semantic violations (g2) do not fall (31 to 41) even as authorization-type violations (g1) collapse (57 to 6). With the deterministic gate (g15), F3 equals F4, the gap is 0.000, and violations are 0.0% at every scale. Scale thus solves capability-type compliance (g1) but not policy-semantic compliance (g2); the gate zeros the residual at every scale, frontier included. This is the policy-enforcement analogue of hallucination inevitability (§2) and the backbone of the two-kind scaffold.

Resolving the completion-rate confound. The absolute g2 count (31 to 41) and the total violation percentage (33.6 to 14.3) are confounded by completion: a higher-pass model completes more trajectories and therefore attempts more writes (526, 683, and 680 writes across 7B, 14B, and 32B), so raw counts conflate ability with exposure. Normalizing to the per-write-opportunity rate removes this confound. The g2 rate per write opportunity is 0.103 (95% Wilson interval [0.080, 0.132]) at 7B, 0.070 ([0.053, 0.092]) at 14B, and 0.075 ([0.058, 0.097]) at 32B: flat across scale, with overlapping confidence intervals and no significant downward trend—if anything the 7B rate is marginally higher. Scale therefore does not reduce the confirm-before-write violation rate, which stays near one in ten to fourteen writes at every scale, while the deterministic gate zeros it everywhere. The strong scale-invariant form thus holds, not merely the differential, though the differential is also clear in that authorization-type violations collapse while policy-semantic violations persist. This range is 7B–32B; the projected 72B point (§7) would extend it. We note that this result currently rests on one violation class (confirm-before-write) on one benchmark, a scope limit we return to in §9.

5.4 Cheap replication: a capability scaffold substitutes for scale

Raw task pass rises monotonically with scale (a single-trial pass rate of 0.24 at 7B, 0.52 at 14B, and 0.60 at 32B, $n = 342$), the "scale buys capability" curve. (This single-trial pass rate comes from a separate scale-decomposition run and is a different estimator from the §5.3 floor-compliance F3 of 0.189 to 0.547, which is robust over three trials; both measure raw task pass and differ by estimator and run, not by contradiction.) The point of the map is that each piece is obtainable more cheaply than scale: a deterministic fetch-first engine—denying ungrounded provenance, calling the producer, and injecting the real value—moves grounding errors from 33 to 9 and roughly doubles pass (from 0.14 to 0.264) with no learning, a capability scaffold standing in for scale. A lever audit shows that authorization, confirmation, ownership, and precondition gates are genuine levers, whereas an eligibility-*steering* gate has essentially no causal effect (the model does not learn to use the guidance) and naive retry is counterproductive.

5.5 The map, filled

function	scale response	genuine limit?	efficient lever
operand execution	saturates by 32B (100% given-spec)	no (at 32B)	base; learning may re-acquire a role at 7B
interference / state load	retire by 32B	no	capability scaffold (present), or scale
length / conditional load	persist at 32B	partial	capability scaffold (partial); conditional → controller or learn
large-candidate selection	flat at all scales	yes (in-head)	deterministic computation
policy compliance (g2)	scale-invariant (per-write rate flat)	yes (guarantee)	guarantee scaffold (gate)
domain facts	n/a	n/a	A2 / retrieval

5.6 Cost: the deployment payoff

On τ^2 -retail, using existing runs, a 32B int8 base with a deterministic gate runs on-premises at approximately \$0.0019 per request versus a frontier API at approximately \$0.044 per request, a ratio of about 23× (and 16–40× over a GPU priced at \$0.2–0.5 per hour); it is roughly 3000× cheaper than a human agent at about \$6 per contact, and it is auditable with zero data egress. The trade-offs are honest: latency is roughly 6× higher (178 seconds versus 30.5), and pure on-premises compliance (0.573) is below the frontier value (0.82), so the claim is near-equal compliance at roughly 23× lower cost, not equal compliance. (The 0.573 figure

is the gated 32B compliance from the deployment run at a single trial, and differs from the 0.547 of §5.3, which is gated 32B compliance over three trials—the same quantity on a different trial count and run. Cost per request is estimated from a token count of approximately characters divided by four together with an assumed GPU utilization; the ratio is robust to an order of magnitude, but the exact multiple awaits measured token accounting, §7.)

A heterogeneous fleet is projected to improve on this: routing easy requests to the 32B on-premises model and hard requests to a frontier model yields a blended compliance of 0.860 (above the pure-frontier 0.816) at \$0.021 per request, roughly $2.1\times$ cheaper, on clean data. We report this as a projection rather than a measurement: it is an arithmetic upper bound that assumes a cheap decidable router. Such a router is feasible because it can reuse the decidable gate and uncertainty mechanism of §5.3 with no new learned component, but it is not yet implemented.

6. Cost-Optimal Lever Allocation and the Knee

The map yields a deployment rule that inverts the "scale it" reflex: reduce the load and supply the guarantee with structure, rather than training the model to tolerate either. A capability scaffold removes the load dimensions that scale would otherwise have to buy, and recedes when scale is available; a guarantee scaffold supplies the property that scale never buys, and is retained at every scale. Per-capability allocation is decidability-first (§3). The A2 is the common adversary and is minimized, while a fixed scaffold and transferable skill are the common ally. The optimum is the cost knee—the smallest total-cost size—and, more generally, a heterogeneous on-premises fleet whose mixture is set by the per-capability crossover between learning and an engine, not by a fixed assumption. Where a load is irreducible by scaffold—conditional depth and path search are our candidates—the lever shifts to a learned, transferable heuristic rather than a larger model. The positioning relative to Palantir Foundry is on cost (hardware, experts, field-crossing, and egress), not on functionality.

7. Future Work

Several cells of the map are measured here and several remain projected; the natural next steps, in order of value, are as follows. The most consequential is a compliance measurement at 72B and above, which would extend the F3/F4 scale-invariance result beyond 32B and is the single largest credibility gain for the central claim. Second, the compliance result should be replicated across additional policy-violation classes and additional benchmarks (SOPBench in particular), since it currently rests on one violation class on one benchmark. Third, controlled multi-turn probes for the state and contradiction load dimensions, and controlled onset curves for length and conditional depth, would complete the load decomposition. Fourth, an independent estimate of ΔL and a test of the load-reduction prediction at two or more scales would move the scaffold-as-load-reduction operator from hypothesis to validated relation. Fifth, a cross-field A2-swap (for example to airline or banking domains) with the fixed component held invariant and no retraining would upgrade transfer from a design premise to a partially measured result. Finally, enabling measured token accounting would turn the cost-per-request figures from estimates into measurements and allow a sweep of size against total cost to locate the knee and the fleet mixture. Items that require frontier delegation or a full multi-field matrix are left beyond the present scope.

8. Discussion

The result reframes the deployment question from model size to lever allocation under cost. Two asymmetries do the work. Capability is scale-retired in a definite order, so a recede-able capability scaffold buys a small model the dimensions that scale would otherwise provide; and compliance is scale-invariant, so a guarantee scaffold is retained even at the frontier. Training a model to tolerate load is training it to do the deterministic controller's job; the design instead routes that work to determinism by default and to learning only for the residual—a residual that is itself scale-dependent, empty for operand execution at 32B but non-empty at 72B, which is precisely why the cost knee, rather than a fixed model size, is the right object.

9. Limitations

Scale range. Our measurements span 7B–32B; 72B, 235B, and frontier scales are projected (§7). The scale-invariance claim is a flat trend supported by priors and would be strengthened by the 72B measurement.

Compliance metric. The strong scale-invariant form is supported by the per-write-opportunity g2 rate (flat across 7B–32B with overlapping confidence intervals, §5.3), not only by the differential; it nonetheless rests on a single violation class (confirm-before-write) on a single benchmark, and cross-class and cross-benchmark replication, together with the 72B point,

remain to be done.

Observational versus controlled evidence. The §5.2 correlations are size-confounded; only the controlled interference probe is causal, and controlled curves for the length and conditional dimensions, and multi-turn probes for state and contradiction, remain to be run.

Cost figures. Cost per request is token-estimated and robust to an order of magnitude but not yet measured; the fleet result assumes a decidable router that is not yet built; and the human and Palantir comparisons are external baselines.

Transfer. Zero-retrain A2-swap is a design premise; cross-field validation is partial, and the underlying "small model plus schema-swap to an unseen domain" mechanism is itself a variant of schema-guided dialogue state tracking and tool-use systems—we claim the scale-invariance map, not the transfer mechanism.

Single benchmark for compliance. The F3/F4 analysis is on τ^2 -retail; cross-domain and cross-benchmark evaluation remain to be done.

10. Conclusion

Scaling a tool-use agent buys capability: it retires load dimensions in a definite order and lifts task pass. It does not buy compliance: policy-semantic violations are flat across 7B to 32B even as pass climbs, and a deterministic gate removes them at every scale. The right object of optimization is therefore not the model size but the capability $\times$ scale $\times$ lever $\times$ cost map—a small base, a capability scaffold that recedes as scale grows, a scale-invariant guarantee scaffold, authored A2, and learning reserved for the scale-dependent residual, assembled at the total-cost knee. This is why even frontier systems should use deterministic scaffolding and authored relations, and it motivates the companion work that generates the A2 and learns the path-selection residual.

References — Paper 1 (verified 2026-06-26)

These are the **only** works cited in the manuscript. `\nocite{*}` was removed from the LaTeX source; the bibliography now contains exactly these 29 entries. Each was opened individually on arXiv and confirmed: exact title, first author, year, and that the in-text use matches the paper's actual content.

(Scratch literature notes `refs_lit.md` / `refs_dr.md` are research-time collections and are **not** the reference list — superseded by this file.)

#	arXiv	First author	Title (verbatim)	Year
1	2207.00729	W. Merrill & A. Sabharwal	The Parallelism Tradeoff: Limitations of Log-Precision Transformers	2022
2	2305.15408	G. Feng	Towards Revealing the Mystery behind Chain of Thought: A Theoretical Perspective	2023
3	2402.12875	Z. Li	Chain of Thought Empowers Transformers to Solve Inherently Serial Problems	2024
4	1911.01547	F. Chollet	On the Measure of Intelligence	2019

5	2401.11817	Z. Xu	Hallucination is Inevitable: An Innate Limitation of Large Language Models	2024
6	2509.04664	A. T. Kalai	Why Language Models Hallucinate	2025
7	2309.02427	T. R. Sumers	Cognitive Architectures for Language Agents	2023
8	2505.07087	R. E. Wray	Applying Cognitive Design Patterns to General LLM Agents	2025
9	2403.00810	F. Zhu	Bootstrapping Cognitive Agents with a Large Language Model	2024
10	2510.09355	F. Yuan	NL2GenSym: Natural Language to Generative Symbolic Rules for SOAR Cognitive Architecture via Large Language Models	2025
11	2402.01817	S. Kambhampati	LLMs Can't Plan, But Can Help Planning in LLM-Modulo Frameworks	2024
12	2211.10435	L. Gao	PAL: Program-aided Language Models	2022
13	2411.15124	N. Lambert	Tülu 3: Pushing Frontiers in Open Language Model Post-Training	2024
14	1807.06096	N. Jansen	Safe Reinforcement Learning via Probabilistic Shields	2018
15	2503.22738	Z. Chen	ShieldAgent: Shielding Agents via Verifiable Safety Policy Reasoning	2025
16	2503.18666	H. Wang	AgentSpec: Customizable Runtime Enforcement for Safe and Reliable LLM Agents	2025
17	2402.00798	Z. Li	Formal-LLM: Integrating Formal Language and Natural Language for Controllable LLM-based Agents	2024

18	2511.21689	H. Su	ToolOrchestra: Elevating Intelligence via Efficient Model and Tool Orchestration	2025
19	2606.06976	Y. Zhou	TRUST: Exploring Agentic Tool-Calling Decisions via Uncertainty-Aligned Reinforcement Learning	2026
20	2604.07036	D. Piatrashyn	ReDACT: Uncertainty-Aware Deferral for LLM Agents	2026
21	2410.10347	J. Dekoninck	A Unified Approach to Routing and Cascading for LLMs	2024
22	2506.07982	V. Barres	τ^2 -Bench: Evaluating Conversational Agents in a Dual-Control Environment	2025
23	2406.12045	S. Yao	τ -bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains	2024
24	2503.08669	Z. Li	SOPBench: Evaluating Language Agents at Following Standard Operating Procedures and Constraints	2025
25	2501.10132	L. Zhong	ComplexFuncBench: Exploring Multi-Step and Constrained Function Calling under Long-Context Scenario	2025
26	2510.12838	Q. Chen	A ² FM: An Adaptive Agent Foundation Model for Tool-Aware Hybrid Reasoning	2025
27	2510.08439	C. Qian	xRouter: Training Cost-Aware LLMs Orchestration System via Reinforcement Learning	2025
28	2504.18851	H. Ross	When2Call: When (not) to Call Tools	2025

29	2605.09252	C.-E. Sun	LLM Agents Already Know When to Call Tools — Even Without Reasoning	2026
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Notes on the verification pass (2026-06-26)

- **Primary-benchmark mis-citation fixed.** The manuscript's primary bench is τ^2 -bench (dual-control); it had been cited as 2406.12045, which is the τ -bench predecessor (Yao et al.). Added the correct τ^2 -Bench entry (2506.07982, Barres et al.) and repointed the citation (τ -bench kept as the named predecessor).
- **TRUST claim corrected.** 2606.06976 ("TRUST", Yijin Zhou) is *uncertainty-aligned RL for tool-calling decisions*, not the "4-way Direct/Tool/Ask/Unable routing, 4B>30B" originally written. In-text rewritten to the paper's actual content.
- **ToolOrchestra numbers confirmed** against the paper's Table 16 (Orchestrator-8B 80.2%@10.3¢ vs GPT-5 77.7%@31.3¢ on τ^2).
- **Reduced from 229 padding entries to the directly-cited set.** `\nocite{*}` removed so the printed bibliography is exactly the cited set — no unverified/fabricated-id exposure.
- **§2.5 named-but-uncited rivals resolved.** Systems named in prose now carry verified citations: A²FM (2510.12838), xRouter (2510.08439), When2Call (2504.18851); the "tool-necessity linearly decodable, AUROC 0.89–0.96" claim is attributed to its real source (2605.09252, Sun et al.). **RTR** and **RITE** could not be found on arXiv (probable fabricated/garbled acronyms) and were **deleted** from the prose. **ARM2** (2510.08163) is real but is a multimodal reasoning-format model, not a router — removed from the routing list. Entry count 25 → 29.